

Survey on Textual Entailment Recognition in NLP

Tommaso Leonardi

Sapienza Università di Roma
leonardi.1914546@
studenti.uniroma1.it

Arianna Paolini

Sapienza Università di Roma
paolini.1943164@
studenti.uniroma1.it

Abstract

Textual Entailment Recognition, also known as Natural Language Inference, is a fundamental problem in Natural Language Understanding, consisting in deciding if a given hypothesis can be reasonably inferred from an input premise.

This paper aims to analyze Natural Language Processing systems abilities in such classification task, which is an important pre-requisite in many downstream applications such as machine translation and question answering.

We review models from the original rule-based systems to current state-of-the-art with deep learning, focusing on the innovations provided by LSTMs and transformer-based models with different attention mechanisms.

We also conduct a comparative analysis on the main evaluation benchmarks (Stanford NLI, Multi NLI, Adversarial NLI and SciTail datasets) of some state-of-the-art models from Google and Microsoft, most of which are BERT-based, finding that they reach approximately 90% accuracy on most datasets but still lack of robustness in adversarial settings presenting diverse text contexts.

1 Textual Entailment Recognition

Textual Entailment Recognition (TER, also known as *Natural Language Inference*) is an application-independent task consisting in automatically recognizing a directional entailment relationship between two text fragments. In particular, given a premise P and an hypothesis H, the goal is to determine whether H is *entailed* by P (which happens when we can say that H is likely true if we assume P to be true).

The first formalization of the TER task [22] defined a binary classification problem where text pairs had to be categorized either as *entailment* or *non-entailment*. Later on, a three-way classification variant of the task was introduced, which is now the standard approach. Text pairs must be recognized as:

- *entailment*, when H logically follows from P;
- *contradiction*, when H is contradicted by P;
- *neutral*, when we can't say if H is true or false just by assuming P.

Some researchers [33] also proposed to think of the relations between premises and hypothesis on a continuous scale, using scalar labels to capture the degree of entailment, instead of having to fall into discrete categories. However, there are few datasets available for this variation of the task yet, so in the following we will focus on the binary and three-way TER variants.

1.1 Examples

We provide some hand-crafted examples for the three-way TER task.

First Example.

Premise: A boy is playing in the garden.

Hypothesis: There is a boy in the garden.

Classification: entailment

Second Example.

Premise: Luna is barking loudly.

Hypothesis: Luna is human.

Classification: contradiction

Third Example.

Premise: Fishing is a common hobby.

Hypothesis: Everyone likes to go fishing.

Classification: neutral

1.2 Real-world applications

The TER task can be seen as an evaluation tool [41] to test the inference capabilities of various NLP systems, which all have in common the need to understand natural language. For example, it is useful for a system to be able to recognize that different semantic forms convey the same meaning, or that a sentence is implied by another, in order to perform tasks such as text summarization or question answering.

The *PASCAL Challenges* [22] introduced TER with the goal to build a broad evaluation framework for *Natural Language Understanding* (NLU) problems. They provided datasets whose examples were obtained by reframing pre-existing data for downstream tasks such as reading comprehension and information extraction. They made the TER task popular by pushing the researchers to achieve better and better performances.

More recent large-scale datasets, such as the *Stanford Natural Language Inference corpus* (SNLI [15]), enabled the evaluation of deep NLP models too, raising interest in the research community. However, given the poor interpretability of such models, quantitative measures of performance such as accuracy on generic datasets are not enough to know which linguistic properties are effectively captured by a model. Poliak et al. [41] thus suggests to leverage datasets which focus on specific semantic phenomena, such as the *Diverse Natural Language Inference Collection* (DNC) [42].

In general, TER finds large application in the reformulation of other NLP tasks. For example, *question answering* can be recasted as TER by crafting different hypotheses as the fusion of the considered question and a candidate answer, and then ranking them by the extent of entailment from the given premise. Similarly, in *text summarization* and *machine translation*, the candidate outputs (summaries or translations) can become TER hypotheses, so to be able to evaluate their quality by means of the degree of entailment from the premise (the original text) [13].

2 Related work

In this section we review the main approaches to the TER task, going from the first rule-based models to deep machine learning systems and current state-of-the-art.

2.1 Rule-based systems

The oldest approach to the TER task, as to NLP in general, is based on models which use simple *if-then* statements to follow a set of rules, which represent domain knowledge, when processing a "set of facts", that is, the input [28].

Such systems analyze linguistic properties of texts and can mainly focus on *lexic*, when dealing with words and their positions, *syntax*, when considering the grammatical structure of sentences, or *semantics*, when involving the study of the meaning of text [13].

2.1.1 Lexical rule-based

Most early TER solutions used lexical-based approaches. They leveraged features such as Bag-Of-Words and N-Grams probabilities to obtain a pair of vectors representing the premise and the hypothesis, which were then compared to measure similarity and classified according to some rules or thresholds [13].

Jijkoun and de Rijke [24], for example, proposed a simple model to tackle RTE which computed a *directed sentence similarity score* between premise and hypothesis (by summing some frequency-weighted similarity scores between words in the two sentences) and then used a threshold (whose value was selected according to experimental results) to decide if the pair contained an entailment or not.

Similarly, Khader et al. [26] used a threshold-based approach to study the suitability of lexical-based TER for Arabic language (which is subject to lexical and syntactic ambiguity), considering the word overlap and bigram matching between premise and hypothesis pairs.

Bar-Haim et al. [14] proposed, instead, an inference framework which leveraged syntactic dependency trees and entailment rules which addressed both generic linguistic structures and lexical-based inferences. An hypothesis h was considered en-

tailed by a premise p if the parse tree for p could be transformed into the parse tree for h by using the inference rules. Such method achieved high precision but low recall, showing that the rules coverage was not enough and that automatic rule learning was maybe needed.

2.1.2 Syntactic rule-based

While lexical approaches were widely used, they didn't take into account the grammatical relations between words. Conversely, syntax-based systems for TER mainly focus on sentence structure similarities between premise and hypothesis.

Alabbas [12], for instance, presented a model using Tree Edit Distance to measure the similarity between pairs of syntactic trees, achieving promising results for the challenging Arabic language.

2.1.3 Semantic rule-based

Semantics is difficult to recognize in an automated way but gives significant insights to NLP systems. Some *logic-based* TER systems tried to formalize semantic relations between linguistic expressions to then be able to classify the relation between premise and hypothesis.

It's the case of the *NatLog system* by MacCartney and Manning [34], which leveraged a theoretic framework where lexical semantic relations across a sequence of atomic edits transforming a premise p into the corresponding hypothesis h were projected to atomic semantic relations (according to the properties of the context) and then joined together to finally decide if $p \models h$ (p entails h), $p \not\models h$ (p contradicts h) or $(p \not\models h) \wedge (p \not\models \neg h)$ (p is neutral to h). This allowed to exploit lexical and syntactic features to identify fundamental semantic relations between sentences like containment and exclusion, while simultaneously avoiding the typical confusion other lexical and syntactic similarity-based systems got when dealing with common linguistic phenomena such as negation and antonymy.

2.2 Machine learning systems

All of the rule-based systems previously mentioned suffer from lack of generalization capabilities, since neglecting a rule will inevitably result in a failure in the context that addresses it. Instead of trying to capture all of the wide range of existing natural language inferences with static

human made rules, researchers started to use machine learning to teach systems to derive them implicitly and automatically.

2.2.1 Shallow approaches

The TER task can be tackled with different ML classification algorithms: some of the most popular are *Logistic Regression*, *Support Vector Machine*, *AdaBoost* and *Decision Trees*. Feature extraction is also needed, in order to convert premise-hypothesis pairs into feature vectors to feed the algorithms; *Bag-Of-Words* and *TF-IDF*, for instance, are often used to measure word overlap between the sentence pairs, but other kinds of information such as named entities, part-of-speech tags, synonyms, antonyms, stems and lemmas can be leveraged too [13].

The TER system proposed by Malakasiotis and Androutsopoulos [35] involved applying ten different string similarity measures (including edit distance, N-gram distance, Cosine similarity, Manhattan and Euclidean distances) to multiple pair of strings, each containing the original words from premise and hypothesis or some extracted information (such as their stems or part-of-speech tags), to then use the results as features for an SVM classifier. The model performed best in the Question Answering subtask of TER.

Fonseca and Aluísio [20] conducted a feature analysis study of their work on TER for the Portuguese language and found that syntactic and word embedding-based similarity features contributed much in the achievement of better results.

More recently, Shajalal et al. [46] presented a threshold-based approach to semantic feature extraction, which filters out sentence representations that could be irrelevant or counterproductive. The system employed an element-wise Manhattan distance vector to capture semantic entailment between premise and hypothesis, along with several other lexical and semantic features. It also leveraged an ensemble of ML algorithms with the majority voting technique, outperforming some known contemporary works.

2.2.2 Deep learning approaches

The leading branch of research in TER (as in most of the current NLP research) is deep learning,

where the burden of feature extraction is managed by the system itself, and the researcher’s main objective is to design an architecture that’s able to capitalize on the dependencies of the data.

There are many approaches in which the community engaged in the past years, the main difference between them being in how the input is treated. A first broad distinction is between *sentence encoding-based* and *matching encoding-based* systems: the former work on encoding the input via sequence modeling, while the latter focus on analysing the matching between premise and hypothesis. [31]

Most of the recent models for TER leverage attention mechanisms for their predictions. A common approach consists in using *intra-sentence attention* (or *inner attention*) to build meaningful representations for the sentences, but it’s also possible to use *cross-sentence attention* between premise and hypothesis to derive important insight on their connections, if any.

While attention is a powerful tool, it needs to be handled with care. Rath and Basak [44] conducted an analysis on Deep Neural Networks for employment in the TER task, starting from a very simple neural network and adding new components one at a time. They achieved better results when employing an LSTM encoder and *ReLu*-based dense layers (plus dropout and normalization layers), whereas they observed slightly worse performance as soon as they added an attention block to the model pipeline.

Many works make use of LSTMs in their architecture, given its ability to choose which pieces of information to retain and which pieces to forget. Rocktaschel et al. [45] achieved state-of-the-art accuracy on SNLI in 2015 using a sentence encoding approach and word-by-word cross-sentence attention: a first LSTM goes through the premise and its memory state is then passed on to a second LSTM that processes the hypothesis, producing a representation h_N of the pair that is then combined with r , an attention-weighted representation of the premise conditioned on the hypothesis, to get the final representation of the pair. They also tried to leverage the attention-weighted representation of the hypothesis conditioned on the premise, but

in their case this didn’t lead to improved performances.

Their work was soon outperformed by Wang and Jiang [49] with a slightly different architecture: in their case, the second LSTM doesn’t use the memory of the first LSTM, and the attention mechanism also considers the hidden states of an *mLSTM* (match-LSTM), a component that models the matching between premise and hypothesis. They also performed an analysis on the word-by-word alignment weights and on the gates of the *mLSTM*, which led them to notice the following:

- the model tries to find correspondences between the subjects in the premise and in the hypothesis, even if the two sentences are describing two different scenes;
- the value for the input gate tends to be lower in correspondence of prepositions and articles;
- the value of the forget gate tends to be low for entailment examples, while on contradiction examples it increases as soon as the model detects a contradicting fact.

An interesting match encoding approach from Zhao et al. [52] involves applying the attention mechanism (structured attention) to binarized trees representing sentences, where each internal node corresponds to a segment of the sentence and each leaf to a single word. Two binary tree-LSTMs process the premise and the hypothesis to compute the alignment from hypothesis tree nodes to premise tree nodes and vice versa, then the entailment relation is calculated recursively starting from the leaves.

State-of-the-art. To this day deep learning approaches achieved the state-of-the-art on the TER task. Recent systems that result in better performances are those involving LLMs with few-shot learning and the fine-tuning of pre-trained transformers.

Wang et al. [50] achieved state-of-the-art performance on SNLI in 2021 with an accuracy of 93.1% by employing RoBERTa-large with *Entailment as Few-shot Learner* (EFL), a framework that improves the performance of few-shot learning by reformulating classical NLP classification tasks as entailment problems. The

labels of the classification tasks are converted into sentences that can then be used as hypotheses, so that a model fine-tuned on TER may be able to treat the task as a TER problem.

In 2019, the state-of-the-art on MultiNLI has been achieved by Jiang et al. [23], with an accuracy of 92%. They developed a framework that improves robustness and efficiency of fine-tuning, SMART, whose main ingredients are *Smoothness-Inducing Adversarial Regularization* and *Bregman Proximal Point Optimization*. As of 2021, anyway, the new state-of-the-art is Microsoft’s Turing NLR v5 XXL, which achieves an accuracy of 92.6%.

In 2023, ANLI’s state-of-the-art performance was achieved by Kavumba et al. [25] thanks to explanation prompting, which was found to improve models robustness to adversarial attacks.

3 Datasets and benchmarks

FraCaS, Cooper et al. 1996. [19]

An hand-built semantic test suite consisting of 346 textual inference problems covering various semantic phenomena. It is the test suite that first allowed the scientific community to tackle the TER problem (even if still not properly formalized), and has been used to evaluate rule-based approaches.

Pascal-RTE challenges, NIST 2005-2011. [1]

The Pascal-RTE challenges, a set of 7 different challenges held from 2005 to 2011, were the most common benchmark before the publication of SNLI, having been cited thousands of times through the years. Every challenge came with a different dataset, each improving on variety, quality and quantity of pairs presented in the previous challenge. RTE-1 had 287 pairs, while RTE-7 provided 21k pairs, allowing researchers to apply machine learning techniques. These challenges hold great value since they encouraged researchers to achieve the best performance, hence enriching the research on TER of many valuable contributions.

SNLI, Bowman et al. 2015. [16]

The Stanford Natural Language Inference Corpus, consisting of 570.152 sentence pairs, was the first dataset enabling the community to work with

deep learning techniques, and is arguably the most prominent dataset for testing TER models. Via Amazon Mechanical Turk, Bowman et al. asked a group of workers to write three hypotheses for each prompted premise (one for each label to force a balanced dataset: entailment, contradiction and neutral). The premises are image captions from the Flickr30k corpus, so each premise actually refers to a particular scene.

MultiNLI, Williams et al. 2017. [51]

The Multi-Genre Natural Language Inference corpus is a dataset of 433k sentence pairs which is modeled after SNLI, but it improves on the variety of text genres, providing multiple text categories on which TER models can be evaluated and trained on. Other than the image captions (already present in SNLI), MultiNLI provides coverage on 9 more genres, as, for example, face to face conversations, governmental statements, telephone speech and letters.

XNLI, Conneau et al. 2018. [18]

The Cross-lingual Natural Language Inference corpus is an extension of the MultiNLI corpus to 15 languages, which aims to be a viable environment for evaluating cross-lingual sentence representations, effectively overlapping with evaluation of machine translation models. The examples are obtained by translating 5.000 test and 2.500 development pairs from MultiNLI in 14 different languages, resulting in a total of 112.5k pairs. Also, by pairing each premise with the corresponding hypotheses in the 15 languages, the number of possible combinations amounts to 1.5M examples.

QNLI, Wang et al. 2018. [48]

The Question-answering NLI dataset is a component of the *General Language Understanding Evaluation* (GLUE) benchmark. It was obtained by repurposing the Stanford Question Answering Dataset [43] for use in the TER task, and it contains 110.4k pairs.

SciTail, Khot et al. 2018. [27]

The Scientific entailment dataset, consisting of 27k examples, was obtained by converting questions from 4th grade and 8th grade exams and from the sciQ dataset to a textual entailment format. The dataset is linguistically challenging

and utilizes a scientific language.

ANLI, Nie et al. 2020. [39]

The Adversarial Natural Language Inference dataset was designed to be challenging to state-of-the-art models as BERT or RoBERTa. The dataset was collected via *Human-And-Model-in-the-Loop Enabled Training* (HAMLET), which mimics adversarial training, but with human annotators acting as the adversary for a model: given a premise, the annotators had to come up with valid hypotheses on which the model fails. The examples are divided into 3 rounds of increasing difficulty, for a total of 169k pairs.

SICK, Marelli et al. 2014. [36]

The Sentences Involving Compositional Knowledge dataset was built with *Compositional Distributional Semantic Models* (CDSMs) in mind, given the lack of adequate benchmarks. It contains 10k pairs that were collected by transforming and enriching captions describing a scene, and each pair has been annotated both in relatedness (a score from 1 to 5) and in entailment.

HANS, McCoy et al. 2019. [37]

The Heuristic Analysis for NLI Systems dataset is a 30k pairs dataset which aims to challenge machine learning models that base their predictions on 3 specific, but common, heuristics. It contains many examples where these heuristics ("lexical overlap", "subsequent" and "consequent") fail, providing 10k examples for each heuristic.

EVALITA TER, Bos et al. 2009. [2]

EVALITA is an evaluation campaign for NLP tasks in the Italian language that has been proposed in various iterations since 2007. In the 2009 edition one of the evaluated tasks was TER, with a dataset of 800 examples.

4 Existing tools, libraries, papers with code

There are many tools online which can be leveraged to train and test TER models. The *Keras SNLI repository* on GitHub [3] exploits the Keras library (an high-level API of the TensorFlow platform) to provide a baseline to train different neural networks on the SNLI dataset, while the *PyTorch NLI repository* [4] contains a code struc-

ture leveraging the PyTorch library for training on different combinations of models and datasets.

Furthermore, papers presenting a TER system often come with a Python implementation: it's the case of *Textual Entailment with Structured Attentions and Composition* by Zhao et al., which is implemented at [5], *Learning Natural Language Inference with LSTM* by Wang and Jiang at [6] and *Recurrent Neural Network-Based Sentence Encoder with Gated Attention for Natural Language Inference* by Chen et al. at [7].

In our comparative evaluation, we use Hugging face *transformers* [8] and *datasets* [9] libraries in a Google Colab environment to efficiently test some pre-trained models from the Hugging Face Model Hub [10] on the most known benchmarks.

5 State-of-the-art evaluation

TER models, like other classification-aimed systems, can be evaluated with the following metrics [44]:

- *accuracy*: it indicates how frequently the model makes a correct prediction

$$accuracy = \frac{\# \text{ correct classifications}}{\# \text{ total classifications}}$$

- *precision*: it reports how many of the examples classified as 'entailment' (or 'neutral' or 'contradiction') by the model were actually labelled as that

$$precision = \frac{\# \text{ true positives}}{\# \text{ true positives} + \# \text{ false positives}}$$

- *recall*: it states which fraction of actual entailments (or neutrals or contradictions) was correctly recognized by the model

$$recall = \frac{\# \text{ true positives}}{\# \text{ true positives} + \# \text{ false negatives}}$$

- *F1 score*: it considers both precision and recall through an harmonic mean (which penalizes extreme values)

$$F1 \text{ score} = 2 * \frac{precision * recall}{precision + recall}$$

However, accuracy is the most popular choice for the models evaluation. In Section 2.2.2 we reported the accuracy scores for some state-of-the-art TER systems.

6 Comparative evaluation

For our comparative evaluation¹, we selected a group of transformer-based systems from Hugging Face model hub [10] which were already fine-tuned on different TER datasets. We chose to evaluate the performance of the following models:

- **MPNet base** [47]: A model from Microsoft that leverages the dependencies among predicted tokens and auxiliary position information to reduce the position discrepancy between pre-training and fine-tuning.

fine-tuned on: SNLI, MNLI.
parameters: 109M

- **ALBERT XXL** [29]: A BERT-based model from Google that leverages two parameter-reduction techniques to lower memory consumption and increase the speed of training.

fine-tuned on: SNLI, MNLI, FEVER-NLI [38], ANLI.
parameters: 223M

- **mDeBERTa-v3 base** [21]: multilingual version of DeBERTa pre-trained on the *CC100 multilingual dataset* (with 100 languages).

fine-tuned on: XNLI, multilingual-NLI-26lang [11].
parameters: 279M

- **ELECTRA** [17]: A model from Google that's trained in an adversarial way: a discriminator tries to detect whether each token in a sentence was replaced by a generator.

fine-tuned on: SNLI, MNLI, FEVER-NLI, ANLI.
parameters: 335M

- **RoBERTa large** [32]: A transformer-based model pre-trained on English language using

a *Masked Language Modeling* (MLM) objective. We include two versions that differ in the fine-tuning data.

fine-tuned on: (1) MNLI. (2) SNLI, MNLI, FEVER-NLI, ANLI.
parameters: 355M

- **DeBERTa-v3 large** [21]: A model from Microsoft that improves on DeBERTa by replacing MLM with *Replaced Token Detection* (RTD).

fine-tuned on: MNLI, FEVER-NLI, ANLI, LingNLI [40], WANLI [30].
parameters: 435M

The datasets on which we perform the evaluation are the following:

- **SNLI**: this dataset provides a good environment for straightforward entailment recognition.
- **MultiNLI**: this dataset contains different kinds of text genres, useful for testing whether a model is good for recognizing entailment in a variety of contexts.
- **ANLI**: the challenge represented by the examples in this dataset makes it a relevant choice for this evaluation.
- **SciTail**: we chose to include a dataset that none of the models were fine-tuned on in order to have a fairer testing environment.

For the evaluation we leveraged Hugging Face libraries (*transformers* and *datasets*) to load the fine-tuned models from the model hub and evaluate them on the datasets. For each model-dataset pair we collected: overall accuracy, precision on all classes, recall on all classes and F1-score on all classes.

6.1 Results

The results of the evaluation are shown in **appendix A**.

Among the selected models, DeBERTa-v3 is the one that achieves the highest performance on most datasets (the only exception being SNLI, for which it wasn't fine-tuned), outperforming the second-best model in accuracy on ANLI-A3, the hardest round of ANLI, with a margin of 10.8% points. This is because of two reasons:

¹the code is found at this link: https://colab.research.google.com/drive/1KcwMKMU-R7zAy2qa6-3XuKQtg3glH3A_?usp=sharing

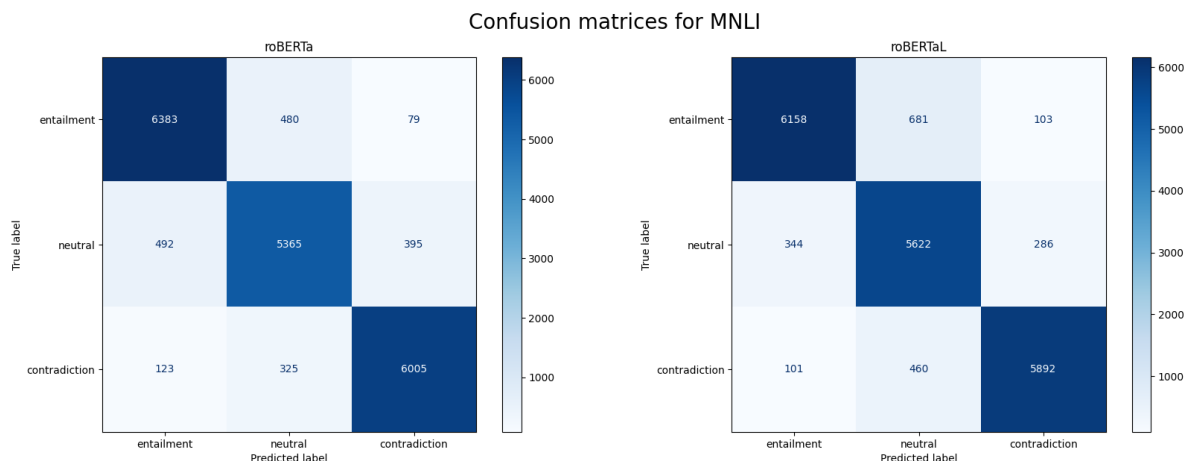


Figure 1: Confusion matrices for RoBERTa(1) (in the picture "roBERTa") and RoBERTa(2) (in the picture "roBERTaL") models in the evaluation on the MNLI dataset

- (1) DeBERTa-v3 has the highest amount of parameters between the models analyzed for this evaluation, allowing it to capture more complex dependencies.
- (2) DeBERTa-v3 was fine-tuned on a more diverse set of datasets compared to other models, having also been trained on LingNLI and WANLI.

By considering the general trend of the results, we can make a few key observations. The first one is that the scores related to *neutral* are generally lower compared to the *entailment* and *contradiction* scores. This means that the models tend to wrongly classify neutral pairs as either contradiction or entailment and vice versa. We can conclude that the neutral class is the hardest to classify since, aside entailment and contradiction pairs being more different from each other compared to neutral ones, neutral pairs can be confusing and ambiguous in some cases (if the premise states that there is *A*, and the hypothesis states that there is *B*, with *A* being a different entity from *B*, should the pair be classified as neutral or as contradiction?). The second observation we can do is that the *contradiction* scores tend to be slightly higher than *entailment* scores, if we exclude ANLI.

By looking more specifically at each dataset, we can say the following:

SNLI: All models fine-tuned on SNLI achieve an accuracy of at least 91%. As for the other models, mDeBERTa has a particu-

larly hard time at correctly classifying contradiction pairs, as the low *contradiction recall* and *neutral precision* scores suggest.

MNLI: On this dataset the models perform almost equally well, with accuracy scores ranging from MPNet’s 87.6% to DeBERTa’s 91%.

ANLI: In all three ANLI datasets DeBERTa is the model that performs best, achieving an accuracy of 79.6%, 68.3% and 64% on the progressively more difficult A1, A2 and A3 rounds.

In ANLI-A1, RoBERTa(2) and ALBERT followed DeBERTa’s leadership quite well with accuracy scores of respectively 73.6% and 72.5%, whereas roBERTa(1) and MPNet struggled with 44.8% and 32.9% respectively.

For ANLI-A2 the situation is similar, with the gap between DeBERTa and other models becoming more evident. RoBERTa(1) and MPNet continue to perform the worst.

In ANLI-A3 RoBERTa(2), mDeBERTa, ALBERT and ELECTRA all stay around 50% accuracy, while RoBERTa(1) and MPNet remain behind with a score of about 25%.

SciTail: All of the models achieve an accuracy of about 80% on SciTail, with DeBERTa performing best with 87.1%.

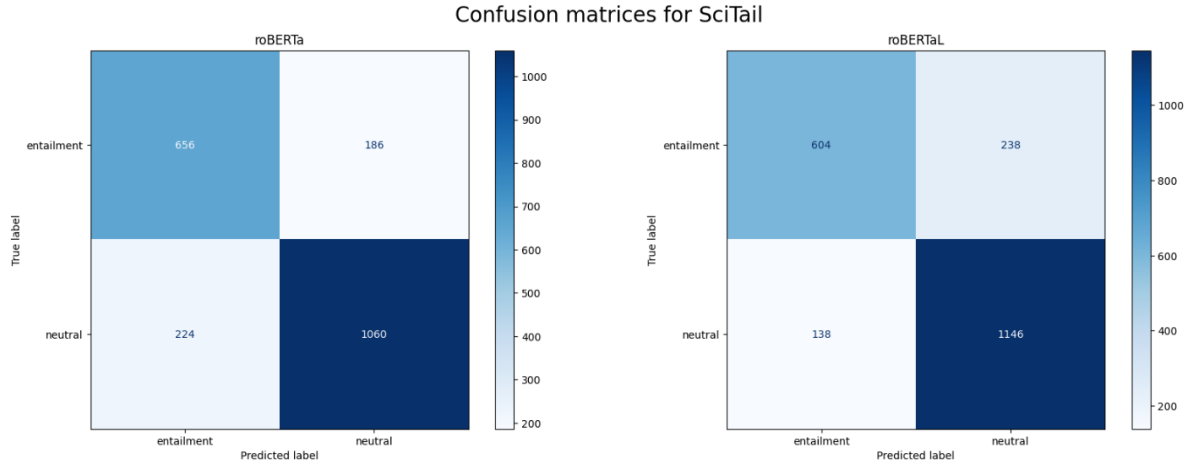


Figure 2: Confusion matrices for RoBERTa(1) (in the picture "roBERTa") and RoBERTa(2) (in the picture "roBERTaL") models in the evaluation on the SciTail dataset

6.2 Discussion

All of the models we analyzed perform worse when it comes to neutral pairs compared to entailment or contradiction pairs. We attribute this problem to an intrinsic challenge of tackling the TER task and to a lack of consensus on what precisely classifies as a neutral pair.

Certainly performing the evaluation without fine-tuning each model specifically on the dataset that it is being evaluated on leads to unfair and misleading results, but at the same time it allowed us to observe how different fine-tuning data influences the performance of the models. An interesting result of this kind is the difference between RoBERTa(1) and RoBERTa(2) on MNLI (Figure 1) and SciTail (Figure 2). We noticed that when these models are evaluated in a fair environment (both are fine-tuned on MNLI and both aren't fine-tuned on SciTail) RoBERTa(1) tends to have higher *entailment recall* and *neutral precision*, while RoBERTa(2) has higher *entailment precision* and *neutral recall*. It looks like a bigger collection of fine-tuning data led RoBERTa(2) to be more strict on entailment classifications and less rigorous on neutral classifications. This could either be a problem caused by limited model capacity, or an effect of training a TER model on a diversified collection of data.

7 Conclusions

In this work, we conducted a survey on the TER task, on the systems that have been developed to

tackle it and on the related datasets that enabled research. We followed the evolution and the variety of the approaches in this field, starting from rule-based systems up until recent deep learning approaches.

After that, we performed a comparative analysis on a series of datasets with a collection of various pre-trained transformer-based models, each fine-tuned on a different set of datasets. We noticed how all of the systems included in the analysis encountered difficulties when classifying neutral pairs, and how fine-tuning of the same model on different collections of data influences the performance.

We think that a more thorough research on what defines a neutral pair in the various datasets might be beneficial to the task, and could also lead to a more rigorous definition of what a neutral pair is, perhaps allowing future models to improve performance on this class.

Another interesting idea for research might be focusing an analysis on how training the same model on different datasets influences the overall performance of a model, which might contribute positively to the development of deployable models.

Contributions.

- Tommaso Leonardi
 - deep learning for TER (Section 2.2.2);
 - datasets and benchmarks (Section 3);
 - comparative evaluation (Section 6);
 - conclusion (Section 7);

- coding and results analysis work on comparative evaluation.
- Arianna Paolini
 - introduction on TER (Section 1);
 - rule-based systems (Section 2.1);
 - shallow ML systems (Section 2.2.1);
 - existing tools, libraries, papers with code (Section 4);
 - state-of-the-art evaluation (Section 5);
 - coding and results analysis work on comparative evaluation.

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A Tables of evaluation results

		precision			recall			F1		
	accuracy	E	N	C	E	N	C	E	N	C
MPNet	91.5%	92.1%	88.1%	94.1%	92.6%	88.3%	93.4%	92.4%	88.2%	93.8%
ALBERT	91.9%	92.4%	86.7%	97%	92.9%	90.8%	91.9%	92.6%	88.7%	94.4%
ELECTRA	91.1%	91%	87.7%	94.6%	91.9%	88.1%	93.3%	91.5%	87.9%	94%
RoBERTa (2)	91.8%	93.2%	86.2%	96.6%	91.8%	91.3%	92.5%	92.5%	88.7%	94.5%
mDeBERTa	76.3%	89%	61.8%	88.5%	80.6%	87.5%	60.5%	84.6%	72.5%	71.9%
DeBERTa	89%	92.7%	80.4%	95.3%	90.8%	89.8%	86.2%	91.7%	84.9%	90.5%
RoBERTa(1)	88.3%	87.8%	84%	93.1%	92.7%	84.1%	87.8%	90.3%	84.0%	90.4%

Table 1: The table of the results on SNLI. The line separates models that were trained on SNLI (above the line) from models that weren't trained on SNLI (below the line). E stands for entailment, N stands for neutral, C stands for contradiction.

		precision			recall			F1		
	accuracy	E	N	C	E	N	C	E	N	C
MPNet	87.6%	90.3%	82.5%	89.8%	88.6%	84.9%	89%	89.5%	83.7%	89.4%
ALBERT	90.2%	92.8%	83.2%	95%	90.2%	89.6%	90.8%	91.5%	86.3%	92.8%
ELECTRA	86.3%	87.1%	83.8%	87.7%	87.5%	82.7%	88.4%	87.3%	83.3%	88%
RoBERTa (1)	90.4%	91.2%	87%	92.7%	91.9%	85.8%	93.1%	91.6%	86.4%	92.9%
RoBERTa (2)	89.9%	93.3%	83.1%	93.8%	88.7%	89.9%	91.3%	90.9%	86.4%	92.5%
DeBERTa	91%	94.2%	84.6%	94.4%	89.8%	90%	93.2%	92%	87.2%	93.8%
mDeBERTa	85.6%	88.9%	77.7%	91%	86.3%	84.9%	85.6%	87.6%	82.2%	88.2%

Table 2: The table of the results on MNLI. The line separates models that were trained on MNLI (above the line) from models that weren't trained on MNLI (below the line). E stands for entailment, N stands for neutral, C stands for contradiction.

		precision			recall			F1		
	accuracy	E	N	C	E	N	C	E	N	C
ALBERT	72.5%	65.2%	75.3%	81.7%	82.9%	76.9%	57.7%	73%	76.1%	67.6%
ELECTRA	67.9%	61.8%	73.4%	70.2%	73.1%	71.2%	59.5%	66.9%	72.3%	64.4%
RoBERTa (2)	73.6%	68.8%	75.5%	77.8%	79.0%	72.4%	69.4%	73.5%	73.9%	73.3%
DeBERTa	79.6%	76.1%	79.8%	83.8%	85.6%	77.2%	76%	80.6%	78.5%	79.7%
MPNet	32.9%	30.3%	39.2%	26.9%	32.3%	45.9%	20.4%	31.3%	42.3%	23.2%
mDeBERTa	64.6%	60.7%	64.7%	70%	68.9%	68.8%	56.2%	64.5%	66.7%	62.3%
RoBERTa (1)	44.8%	43.7%	47.6%	43.5%	54.2%	41.7%	38.4%	48.4%	44.4%	40.8%

Table 3: The table of the results on ANLI-A1. The line separates models that were trained on ANLI (above the line) from models that weren't trained on ANLI (below the line). E stands for entailment, N stands for neutral, C stands for contradiction.

		precision			recall			F1		
	accuracy	E	N	C	E	N	C	E	N	C
ALBERT	59%	57.5%	58.4%	62.9%	72.5%	62.8%	41.7%	64.1%	60.5%	50.2%
ELECTRA	52.2%	51.7%	54.9%	49.8%	57.8%	53.8%	45%	54.6%	54.3%	47.3%
RoBERTa (2)	49.3%	49.1%	50.7%	48.1%	59%	46.2%	42.6%	53.6%	48.4%	45.2%
DeBERTa	68.3%	69.3%	65.1%	70.7%	71.6%	67.3%	66.1%	70.4%	66.2%	68.3%
MPNet	29.8%	30.9%	30%	28.7%	26%	36.3%	27%	28.2%	32.8%	27.8%
mDeBERTa	47.7%	48.1%	48.5%	45.8%	58.1%	52%	33%	52.6%	50.1%	38.4%
RoBERTa (1)	27.4%	30.6%	22.1%	28.8%	32.3%	20.1%	29.7%	31.4%	21%	29.2%

Table 4: The table of the results on ANLI-A2. The line separates models that were trained on ANLI (above the line) from models that weren't trained on ANLI (below the line). E stands for entailment, N stands for neutral, C stands for contradiction.

		precision			recall			F1		
	accuracy	E	N	C	E	N	C	E	N	C
ALBERT	53.2%	56.6%	48.2%	57.8%	58.7%	62.9%	37.6%	57.6%	54.6%	45.6%
ELECTRA	52.3%	54%	48.8%	54.2%	59%	50.5%	47.2%	56.4%	49.6%	50.5%
RoBERTa (2)	45.5%	50.2%	41.4%	45.2%	53.7%	48.5%	34.1%	51.9%	44.7%	38.8%
DeBERTa	64%	67.9%	56.6%	69.4%	66.9%	64.9%	60.1%	67.4%	60.5%	64.4%
MPNet	28.5%	27.9%	29.8%	26.3%	18.7%	47%	19.7%	22.3%	36.5%	22.5%
mDeBERTa	49.8%	52.7%	46.3%	51.2%	55%	56.5%	37.6%	53.8%	50.9%	43.4%
RoBERTa (1)	24.6%	26.7%	23.9%	24.2%	22.6%	31.1%	20.6%	24.5%	27%	22.2%

Table 5: The table of the results on ANLI-A3. The line separates models that were trained on ANLI (above the line) from models that weren't trained on ANLI (below the line). E stands for entailment, N stands for neutral, C stands for contradiction.

		precision		recall		F1	
	accuracy	E	N	E	N	E	N
MPNet	79.2%	73.8%	82.8%	73.8%	82.8%	73.8%	82.8%
ALBERT	82.5%	77.8%	85.6%	78.1%	85.4%	78%	85.5%
mDeBERTa	78.7%	74.8%	81%	69.7%	84.6%	72.2%	82.7%
ELECTRA	79.9%	73.2%	84.7%	77.6%	81.4%	75.3%	83%
RoBERTa (1)	80.7%	74.5%	85.1%	77.9%	82.6%	76.2%	83.8%
RoBERTa (2)	82.3%	81.4%	82.8%	71.7%	89.3%	76.3%	85.9%
DeBERTa	87.1%	85.1%	88.2%	81.6%	90.7%	83.3%	89.4%

Table 6: The table of the results on SciTail. E stands for entailment, N stands for neutral.